# Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 23 repositories, including the decision-chain integration capstone -- one real dataset through the whole distributor chain with 13 machine-checked reconciliation identities plus three additive ones -- an MCP agentic-integration server with contract-enforced result provenance, two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + dispatch study, a visual quality-inspection study with SPC monitoring, a fraud-operations study with gated retrain promotion, a predictive-maintenance policy study, and a live installable quantum-explainer PWA). The euro and accuracy figures throughout are on synthetic data (except retail-analytics-real and decision-chain, measured and labelled on real public data) and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes.

### Job #1 -- (Agentic) Automation with Low-code Platforms

Agentic AI workflows, low-code (n8n / Power Automate), connecting systems & APIs, document automation, ROI, rapid prototyping.

### Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

## Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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# Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

| Area (public-info) | Approach | Repo | Result (synthetic -- labelled if real) |
|---|---|---|---|
| Procurement / assortment / cross-sell | MILP vs greedy; Apriori/FP-growth | revops-optimizer, market-basket-analysis | MILP > greedy; 254 rules, top lift 2.41 |
| Pricing & margin | Elasticity + leakage; endogeneity fix | sales-kpi-analytics, ml-models-lab | EUR2.6M leakage lever, bias +1.52 -> +0.03 |
| Inventory / replenishment | Newsvendor + forecast + ROP/EOQ policy | distributor-intel-platform, ml-models-lab | MASE 0.38; MASE 0.987 / RMSSE 0.948 (beats naive + Holt-Winters); policy: EUR127,421 working capital (cycle+safety exactly), 5.5x turns, 99.9% fill |
| Sales KPIs & analytics | KPIs + rolling-origin CV + pacing interval | sales-kpi-analytics | MASE < 1; exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval -- closed-year back-check landed above the band, reported |
| Logistics / routing | OR-Tools CP-SAT VRP + robustness | route-optimizer | 4.6% / 31% savings; 5% headroom: failing scenarios 96% -> 44%, expected day -4.6% |
| E-procurement automation | Agentic + n8n low-code + seeded reliability benchmark | agentic-automation-lab | ~EUR625k/yr modelled; 1,350 seeded trials: content faults fail silently, stated (assumed rates) |
| Document processing | Agentic extract + 7-rule validation gate | doc-extract-agent | ≈EUR145k/yr modelled; combined-gate precision 70.0% -> 87.5% |
| Customer retention | Decline + churn classifiers | revops-optimizer, ml-models-lab | ROC-AUC 0.99; churn ECE 0.197 -> 0.021; bootstrap skill CI +0.457 [+0.381, +0.528] |
| Warehouse / WMS (new) | WMS twin + factory/plant sim (Story Mode, 894-element plant, definable objects, multi-way line sim + fluids solver); slotting + DES + packing + pick-path routing | logistics-flow-studio (WarehouseTwin v3.19), logistics-digital-twin (engine) | -48.6% pick travel; ABC ~21% > random; ISO 22400 KPIs; 112/112 self-test + 45 harnesses; line sim 112.5 parts/hr (~88.1% eff.); fluids solver (modelled); engine slotting -44.2% (golden-zone 25% -> 100%); fill 2.0% -> 30.2%; routing: return +3.0% vs exact optimum, optimized layout ~46% shorter |
| Supply-network design (new) | Facility MILP + flows + safety stock + service frontier | supply-network-opt | -21.2% cost vs greedy; stock -65.7% / -80.1%; frontier 2,353 -> 14,750 per service point (6.3x) |
| Energy management / facilities (new) | Load forecast (rolling CV) + peak-shaving LP + causal dispatch backtest | energy-demand-forecast | MASE 0.497, 14/14 folds (Holt-Winters loses to naive, reported); peak -20.9%, ~EUR11,100/yr at ASSUMED tariff; battery does not pay back on these assumptions; backtest captures 72.7% of the LP bound (robust variant loses, reported) |
| Quality inspection (new) | Clean-only anomaly detection; pre-registered rule; SPC p-chart + WE rules | quality-anomaly-vision | AE 0.779 vs PCA 0.772 ROC-AUC -- inside 0.02 margin, PCA recommended; TPR 0.407 vs 0.393 @ 5% FPR; calibration correction (ρ) > 0.70 demonstrated |
| Real data (completed) | Cleaning + RFM + returns + leakage-safe CV | retail-analytics-real | real data: seasonal-naive wins CV, MASE 1.094; returns 3.65% of gross, 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0% |
| Chain integration & reconciliation (new) | Provenance-tagged pipeline + identity ledger; MCP agentic layer w/ machine-readable result provenance | decision-chain, chain-mcp | real data, labelled layers: 13/13 identities PASS; additive (p)(q) per-order spread to the cent over 4,151 orders (top decile 59.0%, Gini 0.665); cross-repo revenue GBP 19,643,861.62 to the penny; naive wins lumpy (MASE 1.782); contract-enforced provenance, 145 tests |
| Fraud & transaction-risk ops (new) | Cost-based alerting; champion/challenger gated promotion | fraud-detection-ops | PR-AUC 0.270 vs oracle 0.367; swap-set gates -> PROMOTE at $8,632 vs 8,841$ -- retrain finds no new fraud, wins by shedding load (stated) |
| Maintenance & asset reliability (new) | Censored Weibull + age-replacement vs condition-based | predictive-maintenance | beta 4.81, T* = 44.4 d at 7.16/machine-day -- the calendar rule beats the repo's own detector at the default threshold, reported |

# Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

| Requirement | Repo / artifact | Proof (synthetic unless noted) |
|---|---|---|
| Build AI-agent workflows -- and measure how they fail | agentic-automation-lab (+ eval/reliability.py benchmark) | agentic tool-use loops; reliability benchmark (1,350 seeded trials): content faults fail silently past the guards and retries can't zero them (assumed fault rates, stated); 51 tests |
| Agentic integration via an open standard (MCP) | chain-mcp (official mcp Python SDK; Claude Desktop / Code configs) | 6 real engines as AI-callable tools; typed schemas; never-crash errors; contract-enforced machine-readable result provenance (data label + engine commit + determinism flag); 145 tests incl. live JSON-RPC handshake |
| Low-code (n8n / Power Automate) | agentic-automation-lab: n8n/rfq_intake_agent.json | RFQ-intake agent; ~EUR625k/yr |
| Visual / flow-based building | agent-flow-studio | flow builder; ~EUR47k/yr |
| Connecting systems / APIs | agentic-automation-lab connector layer | agents call external APIs |
| Process automation (back-office) | doc-extract-agent | RFQ/invoice -> structured |
| Document intake / understanding | doc-extract-agent + 7-rule validation layer | extract + validate; ~EUR145k/yr; combined-gate precision 70.0% -> 87.5% |
| Rapid prototyping | agent-flow-studio, agentic-automation-lab, logistics-flow-studio | runnable prototypes incl. a full installable offline WMS twin + factory/plant simulator (PWA, v3.19; 45 harnesses + 112/112 self-test) |
| Prototyping + education (hype-free) | quantum-explainer -- LIVE: dimitres-kisimov.github.io/quantum-explainer | hand-written simulator ~300 lines, zero deps; 42 physics assertions + 57 structural checks pass |
| Show automation ROI | automation-roi-explorer | EUR383k/yr net modelled |
| Python engineering | all automation repos | Python throughout |
| Stakeholder communication | automation-roi-explorer | business-readable ROI views |

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# Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless labelled real data)

| Requirement | Repo / artifact | Proof (synthetic unless noted) |
|---|---|---|
| One dataset through the whole chain (real data + labelled layers) | decision-chain (run artifact + offline dashboard; FAIL path per identity) | 13/13 reconciliation identities PASS + additive (n), (o), (p); cross-repo revenue GBP 19,643,861.62 to the penny; ledger to the cent; forecast-error elasticity == holding share exactly (0.0580); per-order cost-to-serve to the cent over 4,151 real orders -- top decile 59.0%, Gini 0.665 (model shape, not profitability); naive wins lumpy (MASE 1.782); slotting optimum -1.6% vs ABC; CVRP -0.2% vs Clarke-Wright; cost rates labelled INVENTED |
| BI / Power BI | revops-optimizer: powerbi/DAX_measures.md + kpi_uplift_risk | DAX pack + 3-page report + Monte-Carlo P10/P50/P90 risk table w/ DAX |
| KPI dashboards / metrics | sales-kpi-analytics + pacing bullet chart | KPI layer + exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval; back-check: realised EUR11.28M landed above the band -- honest miss |
| Uncertainty / risk quantification | revops-optimizer simulate.py; ml-models-lab BENCHMARK_CI.md | P10-P90 EUR145,091..170,723/yr, ~43% chance of clearing the headline; tornado ranks drivers; bootstrap skill CIs: churn +0.457 [+0.381, +0.528] |
| Inventory policy / working capital | distributor-intelligence-platform dip/inventory.py; supply-network-opt frontier | ROP/EOQ at ABC-XYZ targets: EUR127,421 working capital (cycle+safety exactly), 5.5x turns, 99.9% fill; frontier 6.3x convexity |
| Predictive analytics / forecasting | sales-kpi-analytics; ml-models-lab global forecaster | MASE 0.497 / MAPE 4.8%, 14/14 folds [illegible] |
| Forecasting rigour + reconciliation guard | distributor-intelligence-platform (MRO command center) | MASE 0.38 (9 rolling folds); reconcile guard renders a green 'resilient drift' verdict over 16/16, cross-engine identities with all 21 headline numbers present |
| Energy forecasting + optimization | energy-demand-forecast (forecasters + LP + dispatch backtest; 59 tests) | [illegible] MASE 0.987 / RMSE 0.948 [illegible] peak [illegible] 368.2 -> 291.1 kW (-20.9%), ~EUR11,100/yr at ASSUMED EUR12/kW-month; timer baseline EUR0; battery does not pay back on these assumptions; causal backtest captures 72.7% of the LP bound (robust variant loses) |
| Python for analytics | sales-kpi-analytics, revops-optimizer | end-to-end pipelines |
| SQL / data modelling | sales-kpi-analytics SQL set | spend analysis queries |
| Excel-level tabular analysis | sales-kpi-analytics | spend breakdowns |
| Optimization / prescriptive | revops-optimizer; supply-network-opt; logistics-digital-twin | ~EUR160k/yr; -21.2% cost vs greedy; fill 2.0% -> 30.2% (CP-SAT proof) |
| Elasticity / statistical modelling | revops-optimizer; ml-models-lab | ROC-AUC 0.99; elasticity bias +1.52 -> +0.03; churn ECE 0.197 -> 0.021, bootstrap skill CI +0.457 [+0.381, +0.528] |
| Data into business decisions | sales-kpi-analytics; market-basket-analysis | EUR2.6M leakage lever; cross-sell recs, top lift 2.41 |
| Market-basket / cross-sell | market-basket-analysis + affinity.py | Apriori + FP-growth from scratch; 224 itemsets, 254 rules; affinity network Q=0.58, 3 communities, 4 bridges; 55 tests |
| Classification / anomaly detection | ml-models-lab; fraud-detection-ops | SKU macro-F1 0.963; AE PR-AUC 0.963 vs PCA 0.951 (PCA default); fraud PR-AUC [illegible] (29 tests) |
| Model lifecycle / MLOps (gated retrain) | fraud-detection-ops challenger.py | swap-set + five pre-declared gates -> PROMOTE at 8,632 vs 8,841; retrain finds no new fraud, wins by shedding load ($8/review assumption, stated) |
| Reliability / maintenance (survival) | predictive-maintenance pdm/policy.py | censored Weibull [illegible] age-replacement [illegible] 7.16/machine-day beats the repo's own detector (8.28) at the default threshold -- reported; 63 tests |
| Visual quality inspection (vision) | quality-anomaly-vision (3 detectors, pre-registered rule; SPC; 45 tests) | AE ROC-AUC 0.709 vs PCA 0.772 -- inside 0.02 margin, PCA recommended; TPR 0.407 vs 0.393 @ 5% FPR; texture breaks hard for all (best 0.609); SPC p-chart + WE rules; calibration correction 0 -> 0.70% measured |
| Warehouse / intralogistics + factory / process industry | logistics-flow-studio (WarehouseTwin v3.19); logistics-digital-twin (engine) | WMS twin + factory/plant sim: keyword-generated layout, WMS flow w/ ISO 22400 KPIs, material-flow + KPI dashboard, Safety Mode tour, 8-element signature plant (42 object types), a user-definable object library, multi-way line sim (QA-split, [illegible]) + fluids solver (modelled, not measured), BYO [illegible] random 48.2% pick-travel [illegible] engine slotting -44.2% (golden-zone 25% -> 100%), fill 2.0% -> 30.2%; DES -76.1% / -66.5%; pick-path routing: return +3.0% vs exact optimum, optimized layout ~46% shorter |
| Logistics / ops analytics | route-optimizer (36 tests); supply-network-opt (53 tests) | 4.6% / 31% savings; robustness: 5% headroom cuts failures 96% -> 44%, expected day -4.6%; safety stock -65.7% / -80.1%; frontier 6.3x |
| Real, messy data (completed) | retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0) | real data: 94.0% rows kept, GBP 19.6M revenue; returns 3.65% of gross, 95.0% of value matched, median 10-day lag; Champions 25% -> 69.0%; seasonal-naive wins CV, MASE 1.094 |
| Present to decision-makers | revops-optimizer exec deck | exec narrative deck |